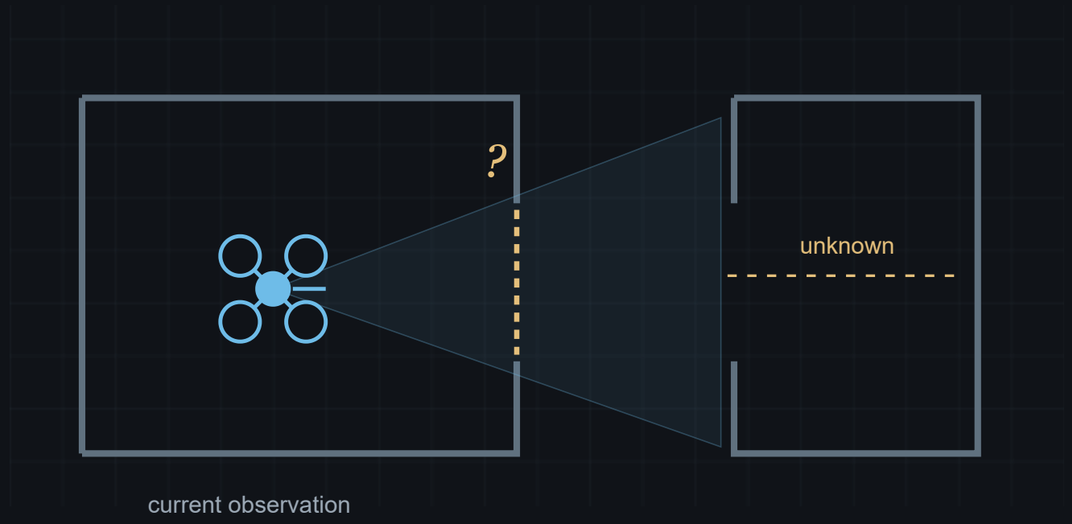
The course is visible.
The next action is uncertain.

Indoor navigation, incomplete observations, and a mission beyond the field of view.



What should happen next?

A passage candidate

What lies beyond it?

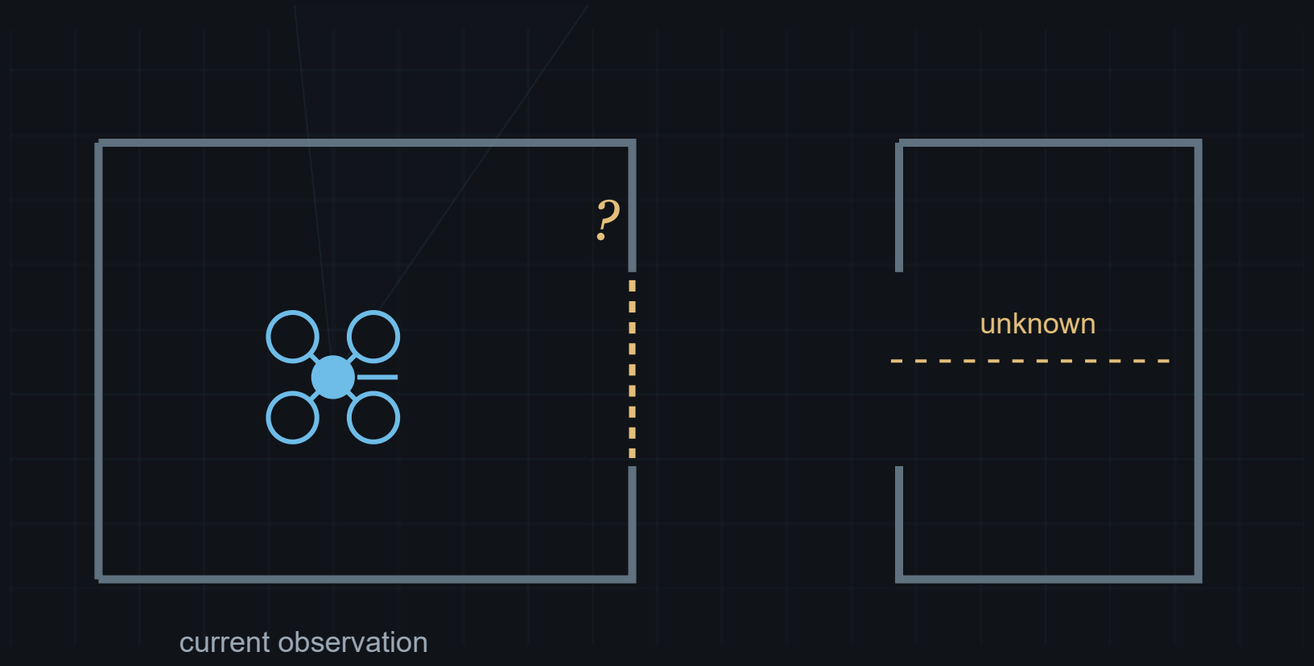
Observe before committing.

geometry + memory + intent

↳ Each camera frame reveals only part of the task.

Build what the drone *remembers*.

Persistent identity. Evidence. Uncertainty. A reason to observe again.



Memory ≠ a fresh view

opening_0003

same entity, new viewpoint

last seen → aging

identity persists · certainty does not

↳ Prioritize a useful world representation before increasing policy complexity.

Illustrative belief update

Combine ideas at the interfaces.

Research gave us the structure; our constraints shaped the implementation.

REPRESENTATION

Hydra

incremental geometry → places

Kimera

metric + semantic structure



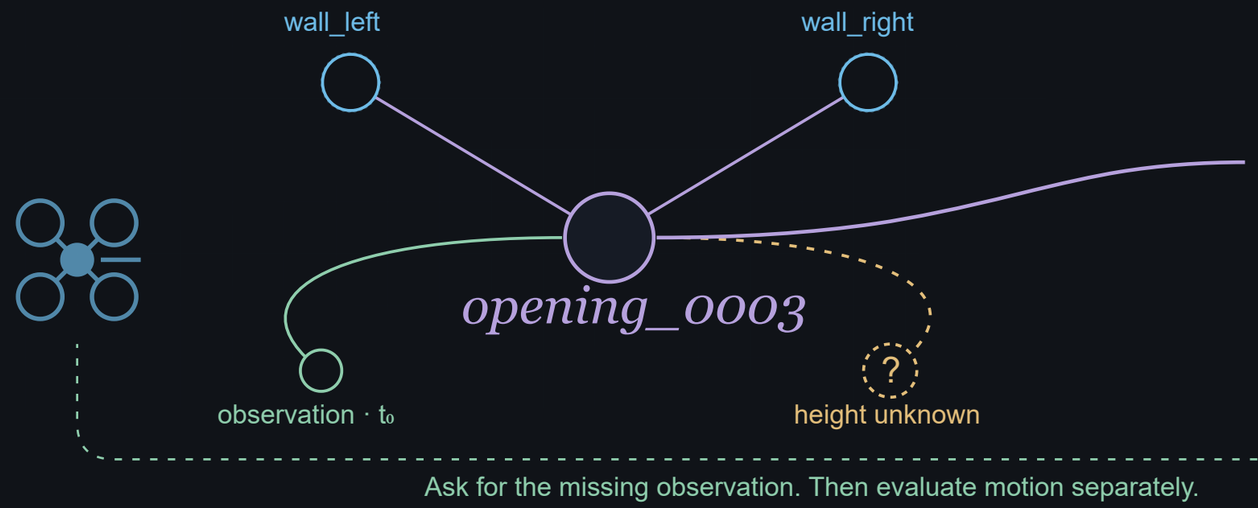
Adapt the ideas. Measure our own system.

↳ A compact adaptation for our task and compute budget.

From seeing to *understanding*.

Observations become geometry, then meaning, then a grounded decision.

01 OBSERVATIONS | 02 GEOMETRY | 03 SEMANTIC GRAPH | 04 REASONING



GOAL PROPOSAL

observe_more

target: opening_0003

because: height is unknown

Grounded proposal ≠ permission to fly

Ask for the missing observation. Then evaluate motion separately.

↳ Missing evidence becomes a reason to look again.

Illustrative entity · offline contract

Sometimes the next action is *to look*.

A bounded reasoning layer turns missing information into an explicit goal.

WHAT WE KNOW

opening_0003

width observed

height unknown

last observation retained

→

Bounded context

relevant entities

evidence + missing facts

→

Goal proposal

observe_more

target: opening_0003

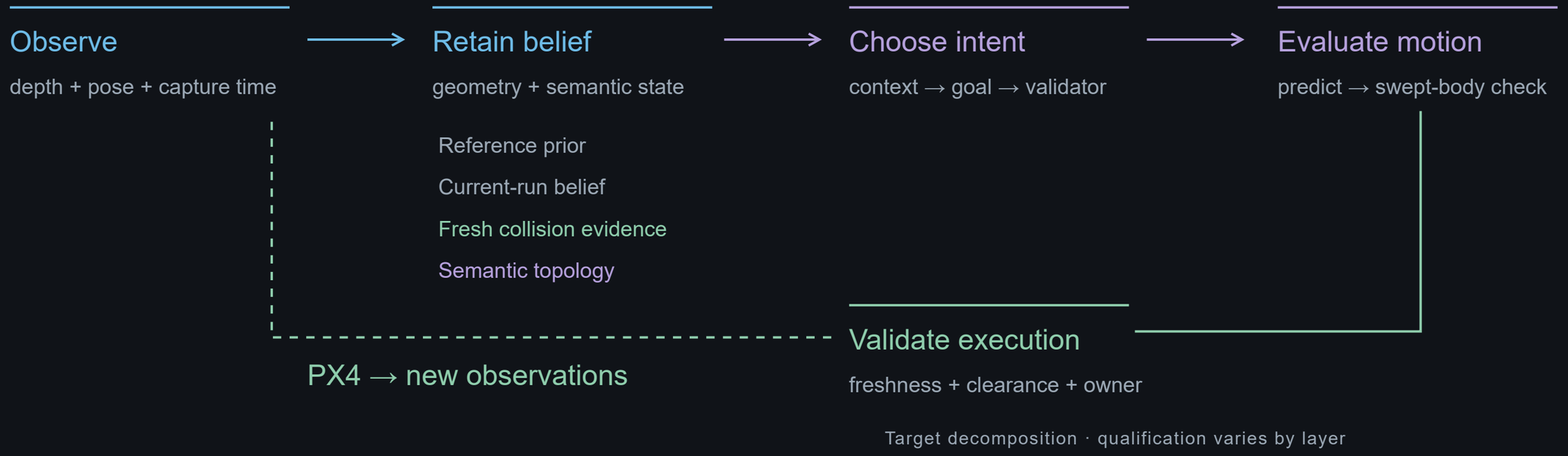
Validate identity, freshness and grounding.

A valid proposal still needs motion evaluation.

↳ “Observe this opening again” is a useful decision.

Each layer answers a different question.

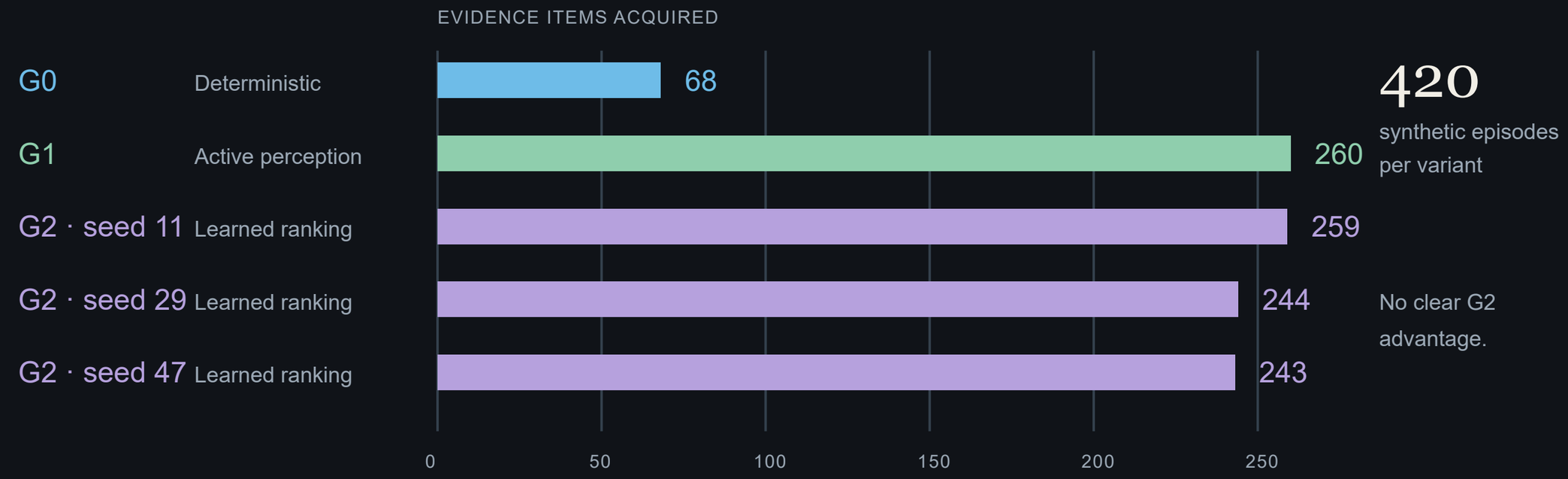
Our target architecture separates memory, intent, motion and execution.



↳ Persistent belief and fresh collision evidence have different jobs.

Complexity has to earn its place.

Active perception earned priority in a synthetic evidence-acquisition task.



420
synthetic episodes per variant

No clear G2 advantage.

↳ Active perception improved this task; learned ranking showed no clear advantage.

Synthetic benchmark · not flight success

A better map was not yet a better decision.

An integration lesson: better estimates must reach the decisions they are meant to change.

EARLY P2 FIXTURE

Better map pose

localization error improved

×

Same stops

planner did not consume it

The missing link was a contract.

Bounded worker → Diagnostic consumer → Decision benefit: unproven

Later offline integration · historical progression

↳ We shifted toward explicit consumer contracts and bounded asynchronous registration.

The assumptions failed between layers.

Native experiments exposed altitude, frame, timing and scheduling contracts.

C1 · NATIVE CALIBRATION

The test never reached braking.

Translate ⟶ Height gate ⟶ LAND

Measurement failed. Landing and disarm completed.

→ Altitude retention

Schematic event sequence · not a telemetry curve

S0 · NATIVE COURSE ATTEMPTS

Rollout ≠ native flight

01 frame origin

02 timestamp skew

03 readiness reset

04 scheduling

Attempts 001–004 stayed disarmed.

→ Frame binding + bounded work

↳ Preserved failures became specific engineering work.

What we built. What we still need to prove.

Evidence is specific to a component, an experiment and an environment.

Offline evidence

- Persistent entities + reasoning
- Synthetic active-perception gain
- Mapping + diagnostic integration

Native evidence

- Disarmed observations
- C0 takeoff / hover / landing
- C1 recovery; measurement failed

Still to demonstrate

- Integrated hybrid course success
- Matched navigation benefit
- Hardware qualification

Separate baseline

Classical S1: one successful Gazebo course run.

↳ The hybrid architecture has not yet demonstrated full-course native success in the fetched record.

From observations to *justified actions*.

The next milestone is a measurable mission benefit from each layer.

01

Reliable native segment

frames · timing · height control

→

02

Matched comparison

same map, start and scoring

→

03

Layer ablations

memory → topology → semantics

Observations → *Grounded belief* → *Justified actions*

Learned extensions only when the evidence warrants them.

↳ Our contribution is a testable path from observations to justified actions.

N1 · N6 · N7 · N8

Planned evaluation